DistilBERT

a distilled version of BERT: smaller, faster, cheaper and lighter

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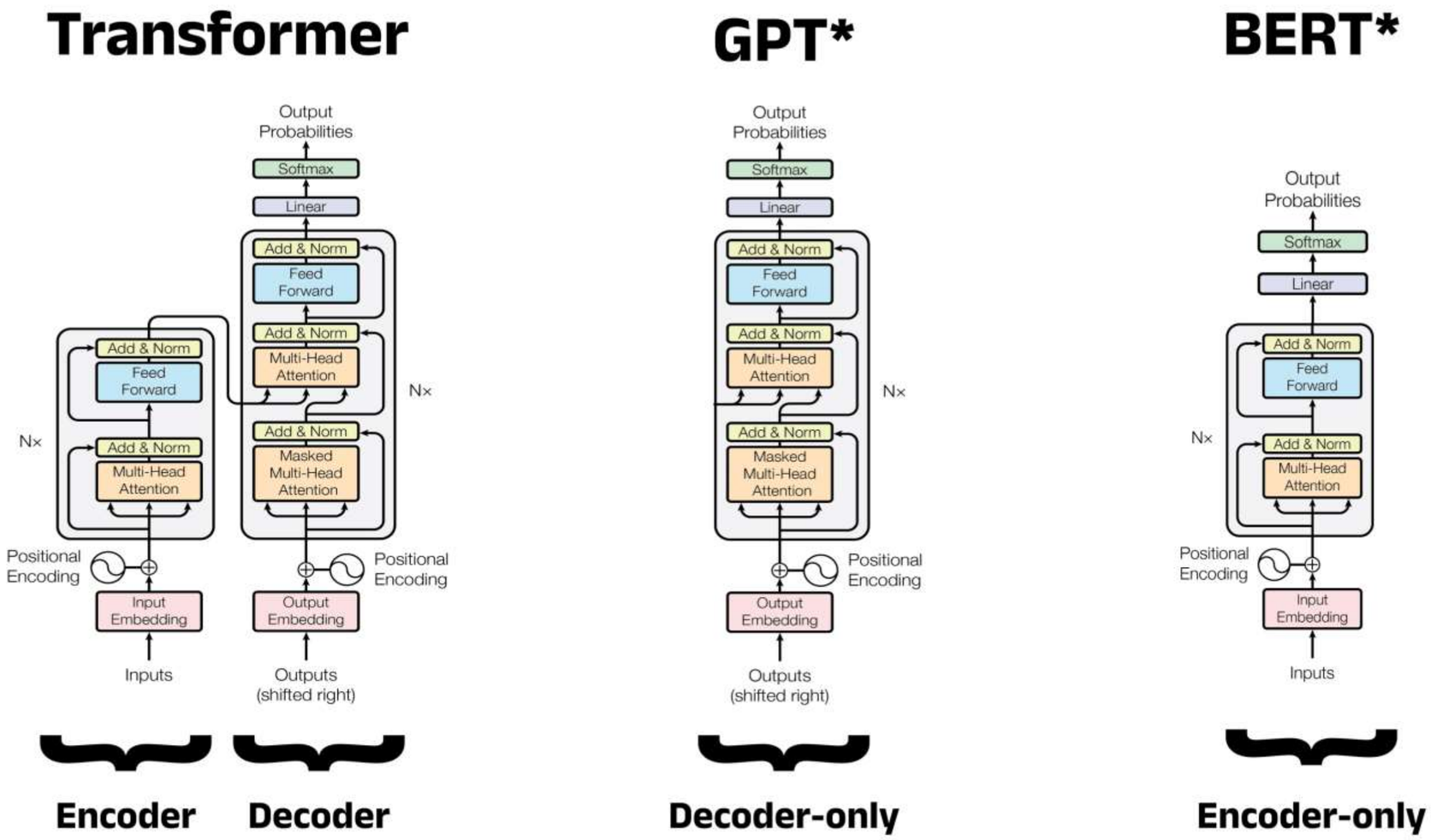
Before starting…

What is BERT?

- **BERT (Bidirectional Encoder Representations from Transformers)** is a pre-trained language model designed to capture deep contextual meaning in text.
- Built entirely on the **Transformer encoder architecture.**
- Relies on self-attention mechanisms to model relationships between words, even when they are far apart.

BERT



*Illustrative example, exact model architecture may vary slightly

Problem Definition

The paper addresses a core tension in modern NLP: performance vs. efficiency.

- Hundreds of millions of parameters (BERT base = 110 million parameters, ≈440 MB for FP32)
- High memory and latency costs at inference time
- Expensive and energy-intensive training
- Limited feasibility for on-device or edge deployment

How can we build a **general-purpose language model** that is significantly smaller and faster than BERT, yet retains most of its language understanding capability across tasks?

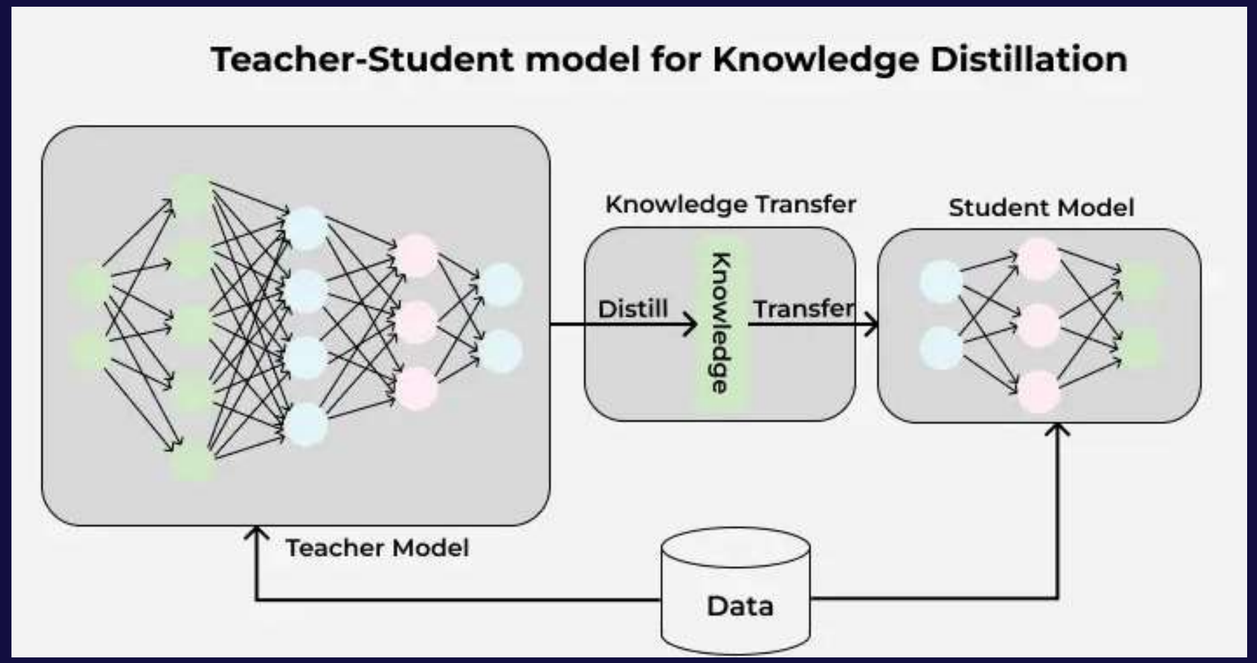
Methodology

Knowledge Distillation Framework

The methodology is based on knowledge distillation, where:

- A **teacher** model (BERT-base) is large and high-performing.
- A **student** model (DistilBERT) is smaller and trained to imitate the teacher.

Instead of learning only from hard labels, the student learns from the soft probability distributions (probabilities for all classes) produced by the teacher, which encode richer information about uncertainty and class similarity.

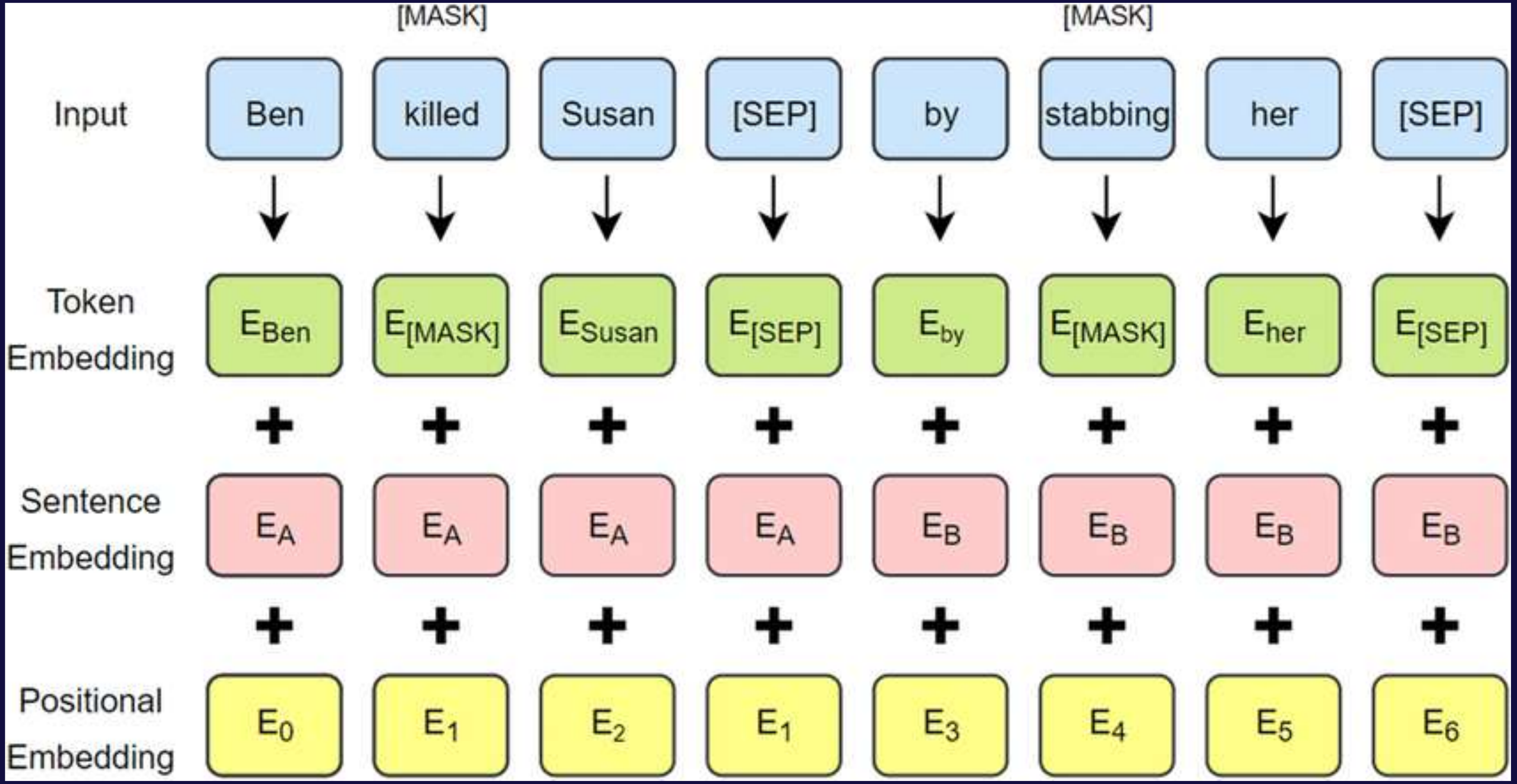


Student Architecture (DistilBERT)

DistilBERT preserves the Transformer encoder design but applies targeted simplifications:

- Same hidden size(768) and attention dimensions(12 heads, 64-dimensional) as BERT(direct weight transfer, copy everything.)
- Number of layers reduced by 50% which reduces computational cost.(6 encoder instead of 12)
- Removal of token-type (segment) embeddings.(which sentence it belongs to)
  - BERT trained with Next Sentence Prediction (NSP)
- Removal of the pooler layer. (used for NSP, sentiment classification, sentence-level decisions)

The authors justify this choice empirically: reducing depth yields larger computational gains than reducing hidden dimensionality for a fixed parameter budget.


[MASK]
[MASK]
Input
Ben
killed
Susan
[SEP]
by
stabbing
her
[SEP]
Token Embedding
E_{Ben}
$E_{[MASK]}$
E_{Susan}
$E_{[SEP]}$
E_{by}
$E_{[MASK]}$
E_{her}
$E_{[SEP]}$
Sentence Embedding
E_A
E_A
E_A
E_A
E_B
E_B
E_B
E_B
Positional Embedding
E_0
E_1
E_2
E_1
E_3
E_4
E_5
E_6

BERT	DistilBERT
Goal: Maximum accuracy	Goal: Efficiency and speed
Number of encoder layers: 12	Number of encoder layers: 6
Number of parameters: ~110M	Number of parameters: ~66M
Token-type(segment) embedding: Used	Token-type(segment) embedding: Removed
Next sentence prediction (NSP): Used	Next sentence prediction (NSP): Removed
Pooler layer: Present	Pooler layer: Removed

Training Objective: Triple Loss

DistilBERT is trained using a **triple loss function**:

- Masked Language Modeling loss (MLM) — standard BERT objective, cross-entropy loss over masked tokens.
- Distillation loss — cross-entropy between student and teacher output distributions
- Cosine embedding loss — aligns student and teacher hidden-state directions

This combination ensures that the student learns:

- Language structure (MLM)
- Teacher decision behavior (distillation loss)
- Representation geometry (cosine loss)

To stabilize training, the student is initialized using **every other layer from the teacher**, leveraging shared dimensionality and reducing optimization difficulty.

Ablation Studies

Ablation experiments isolate the contribution of:

- Individual loss components
- Teacher-based initialization

This allows causal interpretation of what drives performance retention.

Ablation Studies

- L□□□ Masked Language Modeling loss
- Lce = Distillation / Cross-Entropy loss

-Lcos = Cosine loss

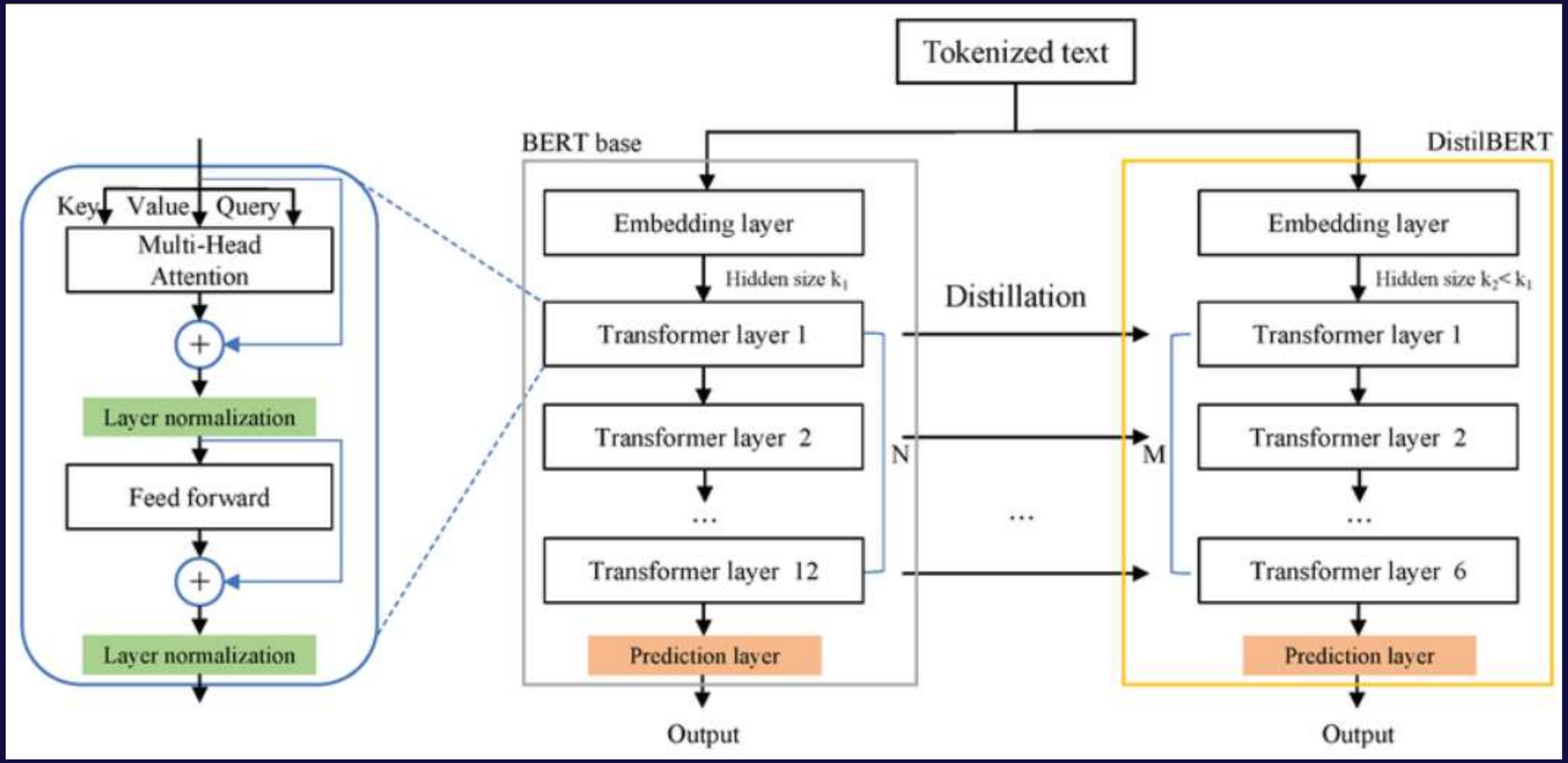
-∅ = not used

Table 4: **Ablation study.** Variations are relative to the model trained with triple loss and teacher weights initialization.

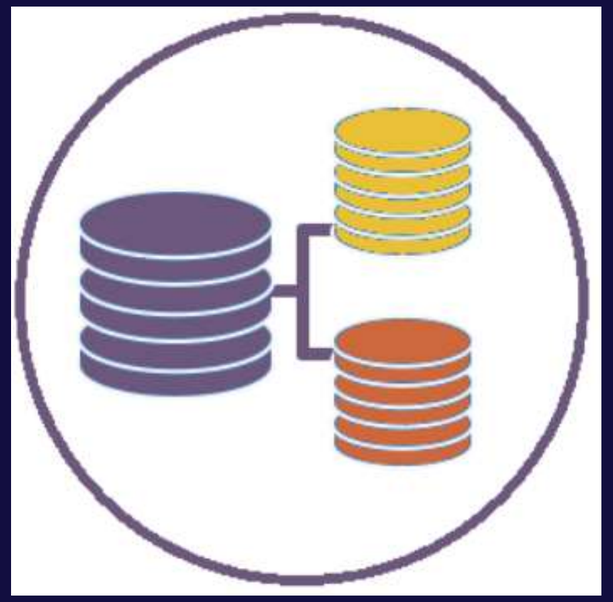
Ablation	**Variation on GLUE macro-score**
$\emptyset$ - L_{cos} - L_{mlm}	-2.96
L_{ce} - $\emptyset$ - L_{mlm}	-1.46
L_{ce} - L_{cos} - $\emptyset$	-0.31
Triple loss + random weights initialization	-3.69

Ablation results show:

- Removing distillation losses causes large performance drops
- Teacher-based initialization is critical for convergence
- MLM loss alone is insufficient to maintain performance


Tokenized text
BERT base
DistilBERT
Key
Value
Query
Multi-Head Attention
Layer normalization
Feed forward
Layer normalization
Embedding layer
Hidden size k_1
Transformer layer 1
Transformer layer 2
...
Transformer layer 12
Prediction layer
Output
Distillation
N
M
...
Embedding layer
Hidden size $k_2 < k_1$
Transformer layer 1
Transformer layer 2
...
Transformer layer 6
Prediction layer
Output

Datasets



3. Pre-training Data

DistilBERT is trained on the **same corpora as BERT**:

- English Wikipedia
- Toronto Book Corpus

This ensures comparability and isolates the effect of distillation rather than data differences.

Evaluation Datasets

The model is evaluated on:

- **GLUE benchmark** (9 NLU tasks)
- **IMDb** sentiment classification
- **SQuAD v1.1** extractive question answering

These datasets test both general language understanding and task-specific reasoning.

Experiments

General Language Understanding (GLUE)

DistilBERT is fine-tuned independently on each GLUE task without ensembling or multi-task training. Performance is reported as the median of 5 runs to reduce variance. Results are compared against:

- ELMo baseline (bidirectional LSTM, Strongly improved many NLP tasks before BERT)
- BERT-base

Downstream Tasks

Additional experiments evaluate performance under realistic inference constraints:

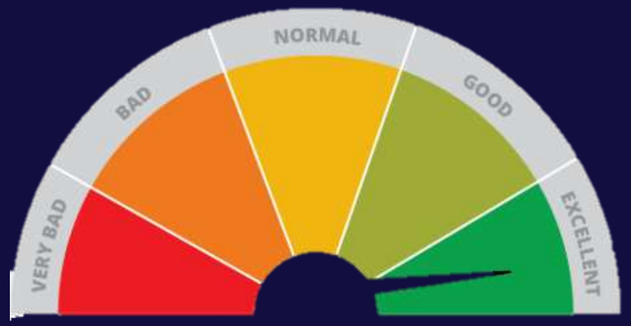
- IMDb sentiment classification (accuracy)
- SQuAD v1.1 (Exact Match and F1)

The authors also experiment with **a second distillation step during fine-tuning** for SQuAD, where a fine-tuned BERT acts as an additional teacher.

Efficiency Evaluation

Efficiency is measured by:

- Parameter count
- CPU inference time on STS-B (Semantic Textual Similarity Benchmark)(batch size = 1)
- On-device inference speed on a smartphone (iPhone 7 Plus)



Results

Table 1: **DistilBERT retains 97% of BERT performance.** Comparison on the dev sets of the GLUE benchmark. ELMo results as reported by the authors. BERT and DistilBERT results are the medians of 5 runs with different seeds.

Model	**Score**	CoLA	MNLI	MRPC	QNLI	QQP	RTE	SST-2	STS-B	WNLI
ELMo	68.7	44.1	68.6	76.6	71.1	86.2	53.4	91.5	70.4	56.3
BERT-base	79.5	56.3	86.7	88.6	91.8	89.6	69.3	92.7	89.0	53.5
DistilBERT	77.0	51.3	82.2	87.5	89.2	88.5	59.9	91.3	86.9	56.3

Across individual tasks (e.g., MNLI, MRPC, QNLI, SST-2), DistilBERT consistently outperforms ELMo and remains close to BERT, demonstrating that pretraining-level knowledge distillation effectively compresses BERT without major loss in language understanding ability.

Results

This table shows that DistilBERT achieves performance close to BERT-base on downstream tasks while being significantly smaller.

Table 2: **DistilBERT yields to comparable performance on downstream tasks.** Comparison on downstream tasks: IMDb (test accuracy) and SQuAD 1.1 (EM/F1 on dev set). D: with a second step of distillation during fine-tuning.

Model	IMDb (acc.)	SQuAD (EM/F1)
BERT-base	93.46	81.2/88.5
DistilBERT	92.82	77.7/85.8
DistilBERT (D)	-	79.1/86.9

On-device inference speed on a iPhone 7 Plus

Table 3: **DistilBERT is significantly smaller while being constantly faster.** Inference time of a full pass of GLUE task STS-B (sentiment analysis) on CPU with a batch size of 1.

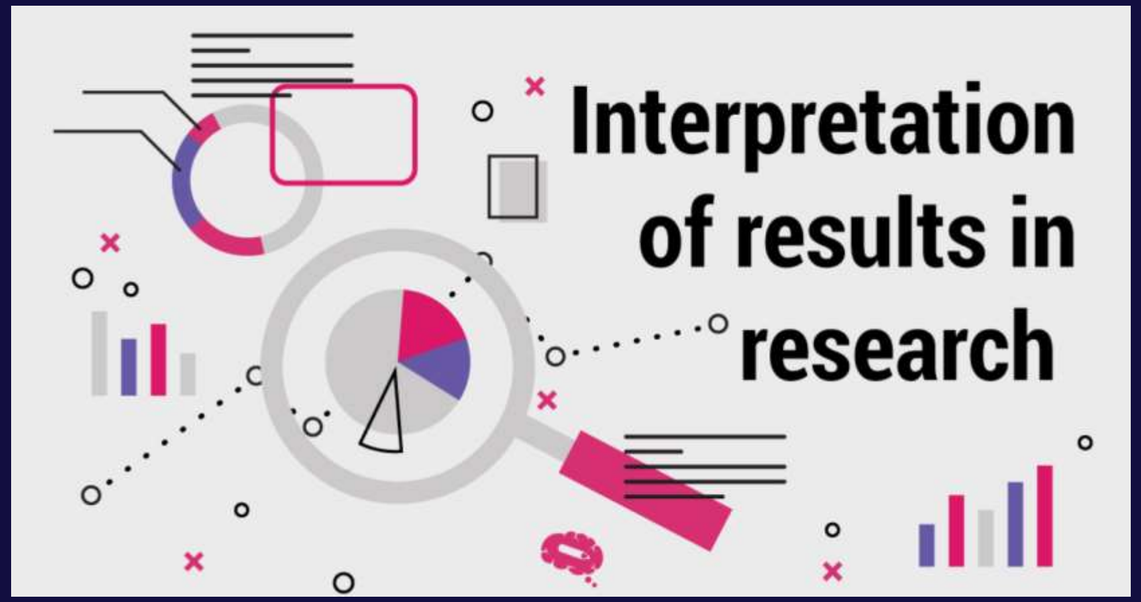
Model	# param. (Millions)	Inf. time (seconds)
ELMo	180	895
BERT-base	110	668
DistilBERT	66	410

Results

Key quantitative findings:

- 40% fewer parameters than BERT (66M vs. 110M)
- ~60% faster inference on CPU
- Retains ~97% of BERT's GLUE performance
- IMDb accuracy within 0.6% of BERT

Interpretation



The results validate the paper's core claim: general-purpose language knowledge can be compressed during pre-training.

Key interpretations:

- Depth reduction is an effective and principled compression strategy.
- Distillation transfers not just predictions, but representational structure.
- Pre-training-level distillation produces models that generalize across tasks.
- Compression trades off some reasoning depth, but gains massive efficiency.

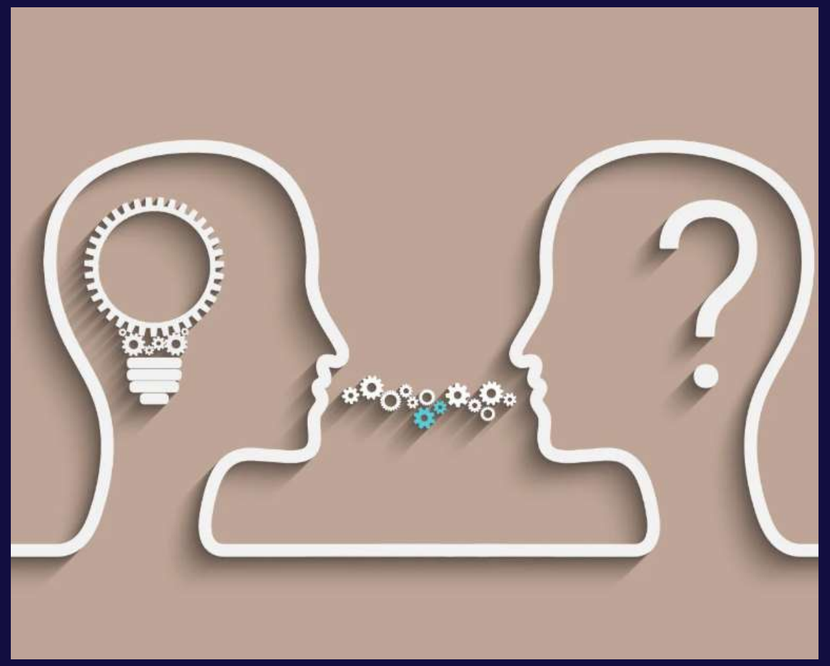
Interpretation

The work also reveals limits:

- Distillation preserves teacher biases. DistilBERT learns by imitating BERT.

- Training remains expensive, Even though DistilBERT is cheap to run. We have to train a BERT model or load a full BERT model.

- Transformer quadratic attention cost remains unsolved which not performs well on long documents. ($O(n^2)$)



Overall, DistilBERT reframes knowledge distillation as a foundational training strategy, not a post-processing trick, and directly influences later compact models designed for real-world deployment.

THANK YOU!